Area Volume and Surface Area

Lesson 12 Choosing Area Volume or Surface Area

Standards 6.G.A.1, 6.G.A.2, 6.G.A.4

Learning target I can choose the correct geometric measure and justify the units.

Materials Rulers, grid paper, unit cubes or paper cube models, paper and scissors for nets.

Lesson Flow 50 Minutes

0–5 Activate prior knowledge. Read the learning target. Ask students what they already know about choosing area volume or surface area. Use a quick model or estimate to make their starting ideas visible.

5–17 Teach and model. Read the requested quantity first and identify the relevant model. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units. Explain why the units and the final value fit the situation. Work through the two examples on the attached instruction page. Have students name the first step before showing it.

17–22 Guided practice. Revisit worked Example 2 with its solution covered. Students propose the next step and explain why it is valid; uncover and compare with the worked solution.

22–45 Independent classwork. Assign the attached 10 questions in order. Check items 1–3 for entry-level understanding, 4–6 for connected procedures, 7–8 for representation and explanation, and 9–10 for application or justification. Ask students to use the instruction-page notes area or extra grids for longer work.

45–50 Check understanding. Use classwork question 10 as an exit discussion: A square pyramid has base edge 14 cm. Each triangular face has perpendicular face height 10 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed. Compare the result with the answer key and ask a student to defend the method.

Support and Extension

Support Model item 1 together with the listed tools, then have students attempt item 2 independently. Watch perpendicular height, missing faces, and square versus cubic units.

Extension Ask students to solve a final item another way, draw a second representation, or create a related problem with a checked answer.

Next step Record which students can explain the target independently. Begin the next teaching session with a brief correction for any recurring error.

Name _____ Date _____

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Learning target I can choose the correct geometric measure and justify the units.

Learn and Discuss

Read the requested quantity first and identify the relevant model. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units. Explain why the units and the final value fit the situation.

Example 1 A rectangular prism has edges 3, 1 1/2, and 6 cm. Sketch it and find its volume.
Solution 27 cm^3. B = 3 × 3/2 = 9/2 cm^2; V = B × 6 = 27 cm^3.
Example 2 A 3 cm by 5 cm base and 4 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.
Solution 79 cm^2. Closed surface area = 2(15+12+20) = 94. Remove the top area 15: 79.

Classwork Show Your Work

Complete the 10 problems on the following page. Refer to the examples, draw models, and check your results.

My Notes or Second Method

Lesson 12 Choosing Area Volume or Surface Area

Name _____ Show calculations, models, and units.

1. Sketch a triangle with base 16 cm and perpendicular height 3 cm. Find its area.

2. A rectangular prism has edges 6, 1, and 5 cm. Sketch it and find its volume.

3. A 6 cm by 2 cm base and 2 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

4. An L-shaped garden is a 11 m by 11 m rectangle with an 7 m by 7 m corner rectangle removed. Sketch and find the garden area.

5. A square pyramid has base edge 8 cm. Each triangular face has perpendicular face height 7 cm. Draw a net and calculate total surface area.

6. Sketch a triangle with base 8 cm and perpendicular height 2 cm. Find its area.

7. A rectangular prism has edges 2 1/2, 2 1/2, and 3 cm. Sketch it and find its volume. Show both lwh and Bh.

8. A 4 cm by 7 cm base and 3 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

9. A 11 m by 7 m rectangle has a right-triangle corner removed with perpendicular legs 3 m and 4 m. Sketch the remaining polygon and find its area.

10. A square pyramid has base edge 14 cm. Each triangular face has perpendicular face height 10 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed.

Choosing Area Volume or Surface Area

Area Volume and Surface Area

1. 24 cm²
Reasoning A = bh/2 = 16 × 3 ÷ 2 = 24. Two such triangles make a parallelogram.

2. 30 cm³
Reasoning B = 6 × 1 = 6 cm²; V = B × 5 = 30 cm³.

3. 56 cm²
Reasoning Closed surface area = 2(12+12+4) = 56.

4. 72 m²
Reasoning Outer area 121 minus cutout 49 = 72.

5. 176 cm²
Reasoning Base = 64; each triangle = 8 × 7 ÷ 2 = 28; add four triangles to get 176.

6. 8 cm²
Reasoning A = bh/2 = 8 × 2 ÷ 2 = 8. Two such triangles make a parallelogram.

7. 75/4 cm³
Reasoning B = 5/2 × 5/2 = 25/4 cm²; V = B × 3 = 75/4 cm³.

8. 94 cm²
Reasoning Closed surface area = 2(28+12+21) = 122. Remove the top area 28: 94.

9. 71 m²
Reasoning Whole rectangle 77; remove triangle 6; difference 71.

10. 476 cm²
Reasoning Base = 196; each triangle = 14 × 10 ÷ 2 = 70; add four triangles to get 476.